

SEGMENTED WORMS

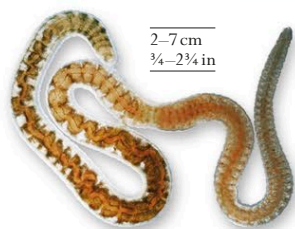
With more complex muscle and organ systems than flatworms, many species of this phylum, called Annelida, are accomplished swimmers or burrowers.

Segmented worms include earthworms, ragworms, and leeches. For most, their blood circulates in vessels and they have a firm sac of fluid (the coelom) running the length of their body, which keeps motion of the gut separate from motion of the body wall. The coelom is split into sections, with each one corresponding to a body segment. Each body segment contains a set of muscles, and coordination of these muscle sets can send a wave of contraction down the length of the body, or make it flex to and fro. This makes many annelids highly mobile both on land and in the water.

Marine segmented worms – the predatory ragworms and their filter-feeding relatives – carry bundles of bristles along their body, typically borne on little paddles that help with swimming, burrowing, or even walking. This group of bristled, segmented worms are called polychaetes or bristle worms.

Land-living earthworms are more sparsely bristled detritus-feeders, and are important recyclers of dead vegetation and aerators of soil. Many leeches are specialized further, and carry suckers to extract blood from a host. Their saliva contains chemicals that stop blood clotting. Other leeches are predatory. Both earthworms and leeches have a saddle-shaped, glandular structure around their body called a clitellum, which they use to make their egg cocoons.

PHYLUM	ANNELIDA
CLASSES	4
ORDERS	17
FAMILIES	About 130
SPECIES	About 18,000



2–7 cm
¾–2¾ in

SLUDGE WORM
Tubifex sp.
F: Naididae

This widely distributed worm can be seen in mud polluted by sewage, the front end buried, the rear end wiggling to extract oxygen.



50 cm
20 in

MEGADRILE EARTHWORM
Glossoscolex sp.
F: Glossoscolecidae

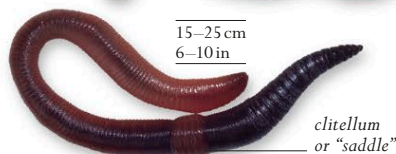
Glossoscolex are large earthworms from tropical C. and S. America. Many are found in rainforest habitats.



10–15 cm
4–6 in

TIGER WORM
Eisenia foetida
F: Lumbricidae

This European inhabitant of rotting vegetation secretes defensive pungent fluid and, like other earthworms, has a “saddle” for producing cocoons of eggs.



15–25 cm
6–10 in

clitellum
or “saddle”

COMMON EARTHWORM
Lumbricus terrestris
F: Lumbricidae

This earthworm – native to Europe but introduced elsewhere – drags leaves into its burrow at night as a source of food.

CHRISTMAS TREE TUBE WORM
Spirobranchus giganteus
F: Serpulidae

Characterized by its spiral whorls of tentacles, used for filter feeding and extracting oxygen, this species is widespread on tropical reefs.

4–7 cm
1½–2¾ in



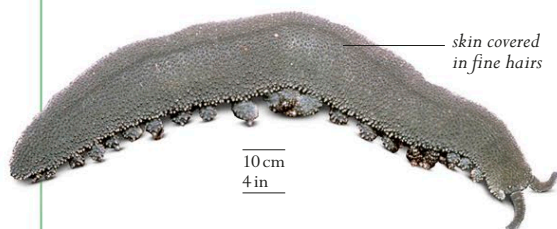
VELVET WORMS

These soft-bodied worms, cousins of arthropods, lumber slowly along dark forest floors like giant caterpillars, but they are extraordinary hunters.

These animals have the body of an earthworm and the multiple limbs of a millipede, but belong in a phylum of their own: Onychophora. Velvet worms are rarely seen in their warm native

rainforests of tropical America, Africa, and Australasia; they shun the open, preferring to hide in crevices and leaf litter. They come out at night or after rainfall to hunt other invertebrates. Velvet worms capture their prey in a unique way: they immobilize their victims by spraying them with sticky slime which is produced by glands that open through pores straddling the mouth.

PHYLUM	ONYCHOPHORA
CLASS	1
ORDER	1
FAMILIES	2
SPECIES	About 200

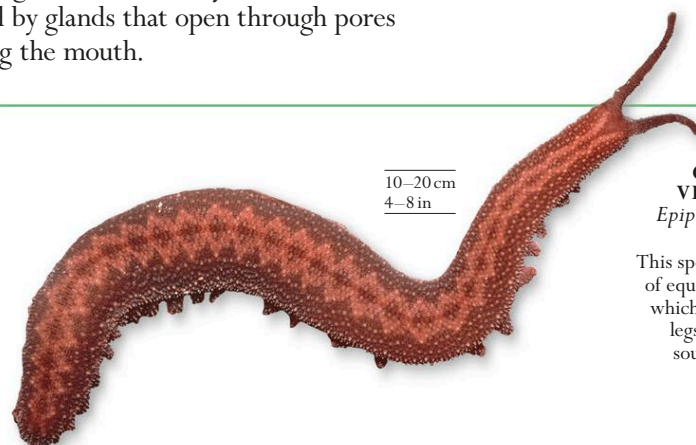


skin covered
in fine hairs

10 cm
4 in

SOUTHERN AFRICAN VELVET WORM
Peripatopsis moseleyi
F: Peripatopsidae

This species belongs to a family with a globally southern distribution.



10–20 cm
4–8 in

CARIBBEAN VELVET WORM
Epiperipatus broadwayi
F: Peripatidae

This species is one of a family of equatorial velvet worms, which typically have more legs than those of the southern continents.